Politecnico di Milano

Biochip A.Y. 2020/2021 - M. Carminati - Exam of 5.07.2021

Test duration: 2 hours – No support material (textbooks, laptops, smartphones, notes) admitted. All the answers must be motivated and the expressions must be provided.

Exercise 1

Consider the following device for DNA analysis including a PCR section and an optical detection section separated by a valve (active with elastic membrane).

a) Find the fluid **speed** in section G with P_{IN} = 2atm P_{OUT} = 1atm in the following **4 cases**: (1) Aux closed, Valve closed, (2) Aux closed, Valve open, (3) Aux = 3atm, Valve closed, (4) Aux = 3atm, Valve open.

b) What are the temperature range to be supported by the PCR section and the accuracy of temperature control system?

c) What solution can be implemented to achieve good mixing of the input samples with primers (Aux) in the PCR chamber?

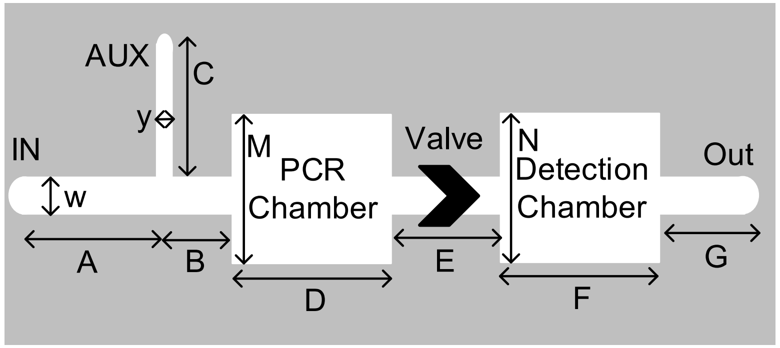
d) What is the impact of **temperature** on diffusion and electrophoretic mobility of molecules?

e) Describe an **on-chip** implementation of a fluorescence analysis system.

f) If the DNA to be analyzed is contained in cells, what techniques can be adopted to extract DNA on chip?

Data:

A = 2 mm
B = 1 mm
C = 2 mm
D = 3 mm
E = 2 mm
F = 2 mm
G = 1 mm
M = 3 mm
N = 3 mm
w = 100 µm
y = 20 µm
1atm = 101 kPa
$\eta_{water} = 10^{-3}$ Pa·s



Channel Thickness h = 20µm

Exercise 2

Consider a cartridge chip for the direct electrochemical quantification of manganese in drinking water (see the voltammogram in Fig.2 for different concentrations in *parts per billion* [ppb = 10^{-9}] in volume).

a) Draw the scheme of the electronics required to perform this electrochemical measurement and size the components and generators (constant and variable) values based on Fig.2.

b) Assuming a max. scan rate of 100V/s, what would be the required bandwidth of the circuits?

c) Choose the optimal potential to operate in **amperometric** mode. What are the limits of amperometry with respect to cyclic voltammetry?

d) Assuming a circular gold WE with diameter of 2mm **in PBS**, what would be the impedance equivalent model of the interface, including the Faradaic term due to manganese? Which parameters can be quantified?

e) Considering only the noise of a 500kΩ feedback resistor, what would be the limit of detection?

f) What drawback is produced by a reduction of V_{DD} (to reduce power dissipation)?

g) What would be the most advisable substrate material on which fabricate the cartridge electrodes? If metal tracks on the cartridge are then covered with passivation, is it better to have a thin or thick top passivation layer?

Fig. 2 - Data: V_{DD} = 5V, V_{SS} = -5V.